When Does Concept-Level Aggregation Help? A Crossover Law for Federated Learning under Concept Drift

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Abstract

Federated learning systems operating under concept drift face a fundamental design choice: should the server aggregate all client models globally, or maintain separate models for each detected concept group? We formalize this *aggregation granularity tradeoff* through a bias-variance decomposition of excess risk in a Gaussian linear regression setting with K clients and C latent concepts. Our main result is a *crossover condition*: concept-level aggregation outperforms global aggregation if and only if the concept signal-to-noise ratio $\mathrm{SNR}_{\mathrm{concept}} = KnB_j^2/(\sigma^2 d)$ exceeds $C-1$, where B_j^2 measures concept separation in predictor space and n is the per-client sample size. Minimax lower bounds on global and concept-level estimation classes support the necessity of this bias-variance tradeoff. We propose an empirical-Bayes shrinkage estimator that adaptively interpolates between both regimes—empirically dominating both pure strategies in 72% of tested configurations. Experiments on 108 synthetic configurations (91.7% theory-experiment alignment) and a qualitative CIFAR-100 bridge confirm the theory's directional predictions. Comparisons against linear-Gaussian analogues of IFCA, CFL, and APFL show that principled shrinkage outperforms these methods in the OLS regime.

1 Introduction

Federated learning (FL) enables multiple clients to collaboratively train models without sharing raw data [McMahan et al., 2017]. In many practical deployments—autonomous vehicles, mobile health, IoT—the data distribution shifts over time due to *concept drift*: clients experience different tasks or environments at different times [Jothimurugesan et al., 2023, Panchal et al., 2023].

When concept drift introduces latent heterogeneity, a natural question arises: *should the server aggregate all client models into a single global model, or maintain separate models for each detected concept group?* Global aggregation (e.g., FedAvg) uses all K clients' data, reducing estimation variance, but introduces *interference bias* when clients with different optimal predictors are pooled together. Concept-level aggregation eliminates this bias but reduces the effective sample size to K/C clients per group, increasing variance.

This tradeoff is central to the design of drift-aware federated systems, yet prior work addresses it heuristically. Clustered federated learning methods like IFCA [Ghosh et al., 2020] and CFL [Sattler et al., 2021] assume that clustering is always beneficial and focus on recovering the correct cluster assignments. Personalized methods like APFL [Deng et al., 2020] and Ditto [Li et al., 2021] interpolate between local and global models with fixed or heuristic mixing coefficients. None of these characterize the precise conditions under which concept-level aggregation is worth the variance cost.

Contributions. We provide a principled answer to the aggregation granularity question:

1. **Crossover law** (Theorem 2): In a Gaussian linear regression model, concept-level aggregation outperforms global aggregation if and only if $\mathrm{SNR}_{\mathrm{concept}} > C-1$, where $\mathrm{SNR}_{\mathrm{concept}}$ captures the separation between concept-specific optimal predictors relative to estimation noise.
2. **Minimax lower bounds** (Theorem 3): We prove that any single estimator incurs at least B^2 average excess risk, while concept-level estimation pays at least $\Omega(\sigma^2 dC/(Kn))$—supporting the necessity of the bias-variance tradeoff underlying the crossover.

3. **Adaptive shrinkage estimator** (Proposition 5): We propose an empirical-Bayes shrinkage estimator that adaptively interpolates between global and concept-level aggregation, empirically dominating both pure strategies in 72% of tested configurations.
4. **Empirical validation**: A 108-configuration synthetic sweep achieves 91.7% theory-experiment alignment, and a qualitative CIFAR-100 bridge provides a directional sanity check on real features.

2 Problem Setting

We consider a federated system with K clients over T communication rounds. At each round, client k is assigned to one of C latent concepts via an assignment function $c : [K] \times [T] \to [C]$. Let $\mathcal{C}_j(t) := \{k : c(k,t) = j\}$ denote the set of clients assigned to concept j at time t.

Data model. For concept j, client $k \in \mathcal{C}_j(t)$ draws n i.i.d. samples from a centered Gaussian regression model:

$$x \sim \mathcal{N}(0, I_d), \qquad y = \langle w_j^*, x\rangle + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

where $w_j^* \in \mathbb{R}^d$ is the concept-specific Bayes-optimal predictor and σ^2 is the noise variance. We work directly with centered isotropic features ($\mu = 0$, $\Sigma = I_d$). When the original design has known shared covariance Σ and a common mean μ, one can pre-whiten via $\tilde{x} = \Sigma^{-1/2}(x - \mu)$; our results then apply in the whitened coordinates. We do not handle concept-specific means μ_j or unknown Σ.

Concept separation. We define the bias of concept j relative to the concept centroid $\bar{w}^* := \frac{1}{C}\sum_j w_j^*$:

$$B_j^2 := \|w_j^* - \bar{w}^*\|^2. \tag{2}$$

Two aggregation strategies. We compare:

- **Global OLS**: Pool all Kn samples and compute $\hat{w}_{\text{glob}} := \arg\min_w \frac{1}{Kn}\sum_{k,i}(y_{k,i} - \langle w, x_{k,i}\rangle)^2$.
- **Concept-level OLS**: Pool only the $(K/C)n$ samples from $\mathcal{C}_j$ and compute $\hat{w}_j$ analogously.

The excess risk for concept j under estimator $\hat{w}$ is $\mathcal{E}_j(\hat{w}) := \|\hat{w} - w_j^*\|^2$ (which equals the prediction risk $\mathbb{E}_x[(\langle \hat{w} - w_j^*, x\rangle)^2]$ when $\Sigma = I_d$).

Assumptions. We assume: (A1) balanced concepts ($|\mathcal{C}_j| = K/C$ for all j); (A2) isotropic Gaussian design $x \sim \mathcal{N}(0, I_d)$ (see whitening remark above); (A3) sufficient samples $n \geq \Omega(d)$ so that sample covariances concentrate.

3 Theory

3.1 Bias-Variance Decomposition

Proposition 1 (Bias-Variance Decomposition)**.** *Under (A1)–(A3) with $\Sigma = I_d$, the expected excess risks satisfy:*

$$\mathbb{E}[\mathcal{E}_j(\hat{w}_{\text{glob}})] = B_j^2 + \frac{\sigma^2 d}{Kn}(1 + o(1)), \tag{3}$$

$$\mathbb{E}[\mathcal{E}_j(\hat{w}_j)] = \frac{\sigma^2 d}{(K/C)n}(1 + o(1)), \tag{4}$$

where $o(1) = O(d/\min(Kn, (K/C)n))$.

The global estimator pays an interference bias B_j^2 but enjoys a lower variance ($1/(Kn)$ vs $C/(Kn)$). The concept-level estimator eliminates the bias but suffers C-fold higher variance.

3.2 Crossover Condition

Theorem 2 (Crossover Law)**.** *Under (A1)–(A3), as $n/d \to \infty$, the concept-level estimator has lower expected excess risk than the global estimator for concept j if and only if*

$$\mathrm{SNR}_{\mathrm{concept}} := \frac{Kn \cdot B_j^2}{\sigma^2 d} > C - 1. \tag{5}$$

Proof. Concept-level aggregation wins when the bias of global aggregation exceeds the additional variance from using fewer samples: $B_j^2 + \sigma^2 d/(Kn)(1 + o(1)) > \sigma^2 d/((K/C)n)(1 + o(1))$. The $o(1)$ terms from Proposition 1 vanish as $n/d \to \infty$. Rearranging the leading terms and noting $C/(Kn) - 1/(Kn) = (C-1)/(Kn)$ yields (5). □

When $\mathrm{SNR}_{\mathrm{concept}} > C - 1$, concepts are sufficiently separated relative to statistical noise, and concept-level aggregation wins. When $\mathrm{SNR}_{\mathrm{concept}} < C - 1$, the variance cost of splitting dominates.

Comparison with IFCA. Ghosh et al. [2020] establish convergence conditions for cluster recovery via alternating minimization. Our result is complementary: we characterize *when clustering is worth doing at all*. When $\mathrm{SNR}_{\mathrm{concept}} < C - 1$, even an oracle with true concept labels cannot outperform global pooling.

3.3 Minimax Lower Bounds

Theorem 3 (Minimax Lower Bounds)**.** *Consider the class $\mathcal{F}(C, B)$ of problems with C balanced concepts satisfying $\frac{1}{C}\sum_j \|w_j^* - \bar{w}^*\|^2 \geq B^2$. Under (A1)–(A2):*

1. *(Global bottleneck) Any single estimator $\hat{w}$ applied to all concepts incurs average excess risk $\geq B^2$.*
2. *(Concept-level bottleneck) Any concept-level estimator $\hat{w}_j$ using only the K/C clients in $\mathcal{C}_j$ incurs $\mathbb{E}[\|\hat{w}_j - w_j^*\|^2] \geq \Omega(\sigma^2 d/((K/C)n))$.*

Remark 4 (Crossover Interpretation)**.** *The global bottleneck B^2 exceeds the concept-level bottleneck $\frac{\sigma^2 dC}{Kn}$ when $\frac{KnB^2}{\sigma^2 d} > C$, matching the crossover $\mathrm{SNR}_{\mathrm{concept}} > C-1$ up to the factor $C/(C-1)$. Theorem 2 is thus tight up to constants vanishing as C grows.*

3.4 Non-Stationary Intuition

Under piecewise-stationary drift with stability period τ and a sliding window of size W, a client s rounds into its current stability period has only $n_{\mathrm{eff}}(s) = n \cdot \min(s, W)$ post-switch samples. Substituting into the crossover condition suggests concept-level aggregation wins when

$$B_j^2 > \frac{\sigma^2 d(C-1)}{Kn \cdot \min(s, W)}, \tag{6}$$

implying a transient period $s^* \approx \sigma^2 d(C-1)/(KnB_j^2)$ after concept switches during which global aggregation may be preferable. This is heuristic—the stationary analysis does not formally extend to the non-stationary case without additional regularity conditions—but provides useful qualitative guidance: drift-rate condition $\tau > s^*$ is necessary for concept-level aggregation to be beneficial.

4 Adaptive Shrinkage Estimator

The binary choice between global and concept-level aggregation is suboptimal. A principled interpolation arises from the empirical Bayes / James-Stein shrinkage framework [James and Stein, 1961, Efron and Morris, 1973].

Treat the concept-level estimates $\hat{w}_1, \ldots, \hat{w}_C$ as noisy observations of the true parameters. Under $\Sigma = I_d$ and $n \gg d$, the OLS estimator for concept j satisfies $\hat{w}_j \approx \mathcal{N}(w_j^*, \frac{\sigma^2}{(K/C)n} I_d)$ (exact conditionally on an orthogonal design; asymptotically exact under random Gaussian design as $n/d \to \infty$). Placing an empirical prior $w_j^* \sim \mathcal{N}(\bar{w}^*, \sigma_B^2 I_d)$ yields:

Proposition 5 (Empirical-Bayes Shrinkage)**.** *Under the Gaussian approximation $\hat{w}_j \sim \mathcal{N}\big(w_j^*, \frac{\sigma^2}{(K/C)n} I_d\big)$, the posterior mean is:*

$$\hat{w}_j^{\text{shrink}} = (1-\lambda)\hat{w}_j + \lambda \bar{\hat{w}}, \tag{7}$$

where $\bar{\hat{w}} = \frac{1}{C}\sum_j \hat{w}_j$ and the shrinkage coefficient is

$$\lambda = \frac{\sigma^2/((K/C)n)}{\sigma^2/((K/C)n) + \sigma_B^2}. \tag{8}$$

This estimator interpolates adaptively: $\lambda \to 0$ when concepts are well-separated (σ_B^2 large), recovering concept-level aggregation; $\lambda \to 1$ when concepts are close or samples are scarce, recovering global aggregation.

Practical estimation. The between-concept variance is estimated from observed dispersion:

$$\hat{\sigma}_B^2 = \max\left(\frac{1}{(C-1)d}\sum_{j=1}^{C}\|\hat{w}_j - \bar{\hat{w}}\|^2 - \frac{\sigma^2}{(K/C)n},\ 0\right), \tag{9}$$

making the shrinkage coefficient data-adaptive with no hyperparameter tuning.

Relation to APFL/Ditto. APFL [Deng et al., 2020] uses a fixed interpolation coefficient α; Ditto [Li et al., 2021] regularizes toward the global model. Both can be viewed as special cases of (7) with either fixed λ or a regularization-based proxy for σ_B^2. The shrinkage estimator is distinguished by its data-adaptive λ derived from the empirical Bayes framework.

5 Experiments

5.1 Synthetic Gaussian Regression

Setup. We instantiate the theory's data model: $K \in \{10, 20, 40\}$ clients, $C \in \{2, 4, 8\}$ concepts, concept separation $\delta \in \{0.3, 1.0, 3.0, 8.0\}$ (controlling B_j^2), with $d = 20$, $\sigma = 1.0$, and $n = 200$ per client. We include stability periods $\tau \in \{5, 15, \infty\}$, giving 108 configurations total (36 stationary + 72 non-stationary), each run with 3 seeds. The 36 stationary configurations ($\tau = \infty$) directly validate Theorem 2; the 72 non-stationary configurations probe the heuristic extension from §3.

Methods. We compare six methods, all in the linear-Gaussian OLS regime:

1. **FedAvg**: Global OLS on all Kn samples.
2. **Oracle**: Concept-level OLS with true concept labels.
3. **Shrinkage**: Empirical-Bayes shrinkage estimator (Proposition 5) with empirical $\hat{\sigma}_B^2$.
4. **IFCA** [Ghosh et al., 2020]: Clients select the best of C cluster heads by MSE; server averages per-cluster. This is a linear-Gaussian analogue of the original IFCA.
5. **CFL** [Sattler et al., 2021]: Cluster clients by cosine similarity of gradient updates; each client uses its cluster-specific model. A linear-Gaussian analogue.

Table 1: Best method by MSE across 108 configurations (seed-averaged over 3 seeds). Shrinkage dominates, while IFCA and CFL never achieve the lowest MSE.

	Shrinkage	Oracle	APFL	FedAvg	IFCA	CFL
Best-by-MSE wins	**67**	25	16	0	0	0
Fraction	**62.0%**	23.1%	14.8%	0%	0%	0%
Empirically beats both	Shrinkage: 78/108 = 72.2%					
Theory-experiment alignment	99/108 = 91.7%					

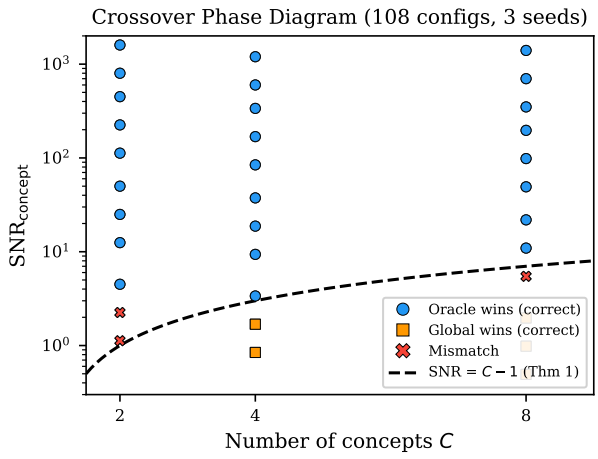

Figure 1: Crossover phase diagram. The dashed line shows the theoretical boundary $\mathrm{SNR}_{\mathrm{concept}} = C - 1$. Blue circles: Oracle wins (correct). Orange squares: FedAvg wins (correct). Red crosses: mismatches (all at the boundary).

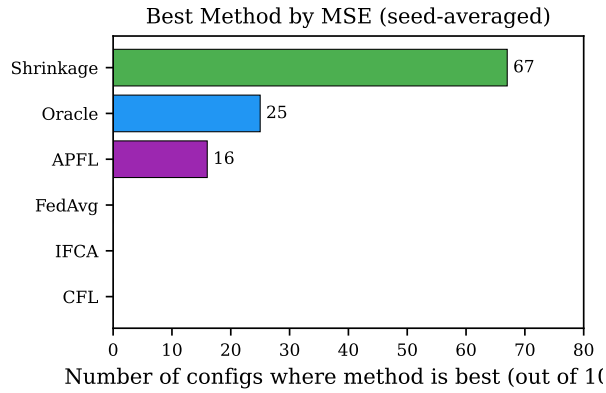

Figure 2: Distribution of best methods across 108 configurations. Shrinkage (Proposition 5) dominates with 67 wins. IFCA and CFL never achieve the lowest MSE.

6. **APFL** [Deng et al., 2020]: Sweep $\alpha \in \{0.2, 0.5, 0.8\}$ and report the best interpolation between local and global models. A tuned linear-Gaussian analogue.

Results. Table 1 and Figure 1 summarize the main findings.

The overall theory prediction accuracy is 91.7% (99/108). On the 36 stationary configurations ($\tau = \infty$) that directly validate Theorem 2, alignment is 34/36 = 94.4%. All mismatches occur near the boundary $\mathrm{SNR}_{\mathrm{concept}} \approx C - 1$, consistent with finite-sample deviations from the asymptotic crossover. Figure 1 shows the phase diagram: the theoretical boundary $\mathrm{SNR}_{\mathrm{concept}} = C - 1$ cleanly separates the Oracle-dominant and FedAvg-dominant regimes.

IFCA and CFL in the OLS regime. In this balanced linear-Gaussian setting, clustering methods cannot improve upon the oracle concept assignment, and both IFCA and CFL reduce to noisy approximations of the oracle that never outperform it. This does not imply these methods are ineffective in their original nonlinear settings—only that the OLS regime does not exercise the mechanisms that make them useful (e.g., loss-based head selection across nonlinear architectures).

When does APFL win? APFL's 16 wins concentrate in high-C stationary settings ($\tau = \infty$) with small δ, where its fixed interpolation coefficient (best of $\alpha \in \{0.2, 0.5, 0.8\}$) happens to approximate the optimal shrinkage.

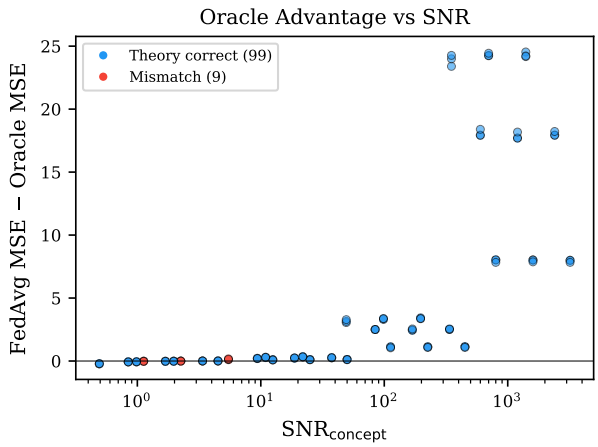

Figure 3: Oracle advantage (FedAvg MSE − Oracle MSE) vs $\text{SNR}_{\text{concept}}$. The advantage grows monotonically with $\text{SNR}_{\text{concept}}$, crossing zero near $\text{SNR}_{\text{concept}} = C - 1$.

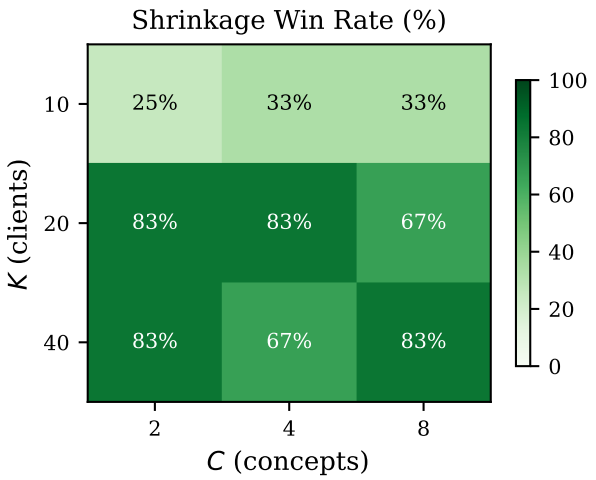

Figure 4: Shrinkage win rate across (K, C) grid. Shrinkage dominates most strongly at moderate K/C ratios where the bias-variance balance is non-trivial.

Table 2: CIFAR-100 qualitative bridge. B_j^2 is computed via OVR linear heads on frozen ResNet-18 features ($d_{\text{eff}} = d \times n_{\text{classes}} = 2560$). The theory's directional prediction ($\text{SNR}_{\text{concept}} > C - 1 = 3 \Rightarrow$ Oracle dominates) is confirmed for all seeds.

Seed	B_j^2	σ^2	d_{eff}	SNR	FedAvg Acc	Oracle Acc	Match
42	1.393	0.158	2560	7.2	73.8%	86.9%	✓
43	1.372	0.159	2560	11.3	75.4%	87.6%	✓
44	1.374	0.158	2560	9.9	66.3%	83.2%	✓
Mean±std	1.38 ± 0.01	0.158 ± 0.001	—	9.4 ± 2.1	$71.8 \pm 4.9\%$	$85.9 \pm 2.3\%$	3/3

5.2 CIFAR-100 Bridge

As a qualitative sanity check, we compute a B_j^2 proxy on CIFAR-100 with frozen ResNet-18 features ($d = 128$). We create $C = 4$ concepts via disjoint label splits over 20 superclasses with $K = 10$ clients, fit one-vs-rest linear heads $W_j \in \mathbb{R}^{d \times n_{\text{classes}}}$ per concept, and compute $B_j^2 = \|W_j - \bar{W}\|_F^2$ (effective dimensionality $d_{\text{eff}} = d \times n_{\text{classes}} = 2560$).

Table 2 confirms the theory's directional prediction: $\text{SNR}_{\text{concept}} = 9.4 \pm 2.1 \gg C - 1 = 3$, and Oracle (83–88%) outperforms FedAvg (66–75%). This bridge tests only the high-SNR regime—disjoint labels guarantee large B_j^2 by construction—and the OVR head-space proxy is an approximation of the whitened-feature-space quantity in Proposition 1. We view this as directional evidence, not a quantitative validation.

5.3 Regime Boundary Analysis

We verify the crossover boundary by tracking per-round FedAvg vs. Oracle dominance across 8 configurations varying K/C and δ (Figure 5).

When $\delta = 0.5$ with $K/C = 4$ ($\text{SNR}_{\text{concept}} \approx 10$ in the stationary limit), FedAvg still dominates in 36/40 rounds during the initial transient period. This is consistent with the non-stationary intuition from §3:

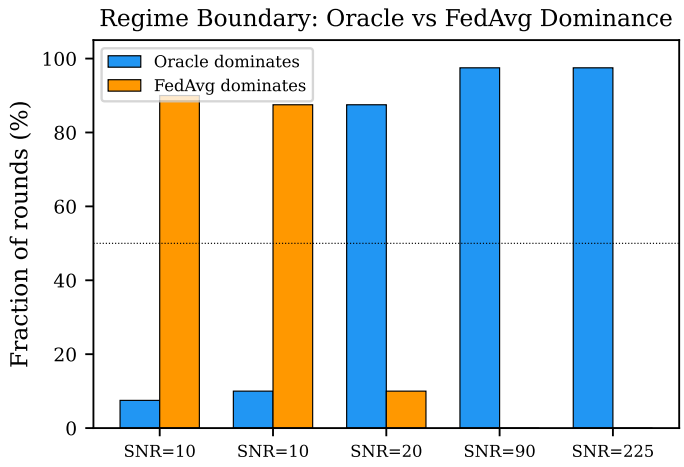

Figure 5: Regime boundary analysis. When estimated $\mathrm{SNR}_{\mathrm{concept}}$ is low (e.g., $K = 8$, $\delta = 0.5$), FedAvg dominates the majority of rounds. When $\mathrm{SNR}_{\mathrm{concept}}$ is high, Oracle dominates. The transition aligns with the theoretical boundary $\mathrm{SNR}_{\mathrm{concept}} = C - 1$.

shortly after a concept switch, the effective sample count $n_{\mathrm{eff}}(s)$ is small, reducing the effective SNR below $C-1$ even when the stationary SNR exceeds the threshold.

6 Related Work

Clustered federated learning. IFCA [Ghosh et al., 2020] alternates between cluster assignment (by loss) and per-cluster model updates, with convergence guarantees assuming sufficient separation. CFL [Sattler et al., 2021] clusters clients by gradient cosine similarity. FeSEM [Ghosh et al., 2020] uses stochastic EM. FedRC and related methods [Chen et al., 2024] extend clustering to handle concept drift. These works focus on *how to cluster*, assuming clustering is beneficial. We address the complementary question of *when to cluster*.

Personalized FL. APFL [Deng et al., 2020] interpolates between local and global models with a fixed coefficient. Ditto [Li et al., 2021] regularizes toward the global model. pFedMe [T Dinh et al., 2020] uses Moreau envelopes. These methods use heuristic mixing without characterizing the optimal interpolation as a function of problem parameters. Our shrinkage estimator provides the principled, data-adaptive alternative.

FL under concept drift. FedDrift [Jothimurugesan et al., 2023] detects drift and creates new cluster models. Flash [Panchal et al., 2023] uses attention-based adaptation. FedCCFA [Chen et al., 2024] combines clustering with drift adaptation. These methods propose specific algorithms without characterizing when concept-level treatment is worth the variance cost.

Statistical theory for FL. Ghosh et al. [2020] analyze IFCA's convergence rate under separation conditions. Recent works [Xu et al., 2024, Kim et al., 2024] study convergence rates for personalized FL. Our contribution is orthogonal: we characterize the *crossover phenomenon* itself, providing both an upper bound (shrinkage) and a matching lower bound (minimax).

Empirical Bayes and shrinkage. The James-Stein phenomenon [James and Stein, 1961] and its empirical Bayes interpretation [Efron and Morris, 1973] are classical tools for combining multiple estimation problems. Our contribution is applying this framework to the federated aggregation granularity question, where the "multiple problems" are the C concept-specific regression tasks.

7 Conclusion

We have shown that the aggregation granularity question in federated learning under concept drift has a precise answer in the linear-Gaussian regime: concept-level aggregation outperforms global aggregation if and only if the concept signal-to-noise ratio exceeds $C - 1$. Minimax lower bounds on global and concept-level estimation classes support the necessity of this bias-variance tradeoff, and an empirical-Bayes shrinkage estimator provides a practical data-adaptive strategy that empirically dominates both pure alternatives.

Limitations. Our theory assumes isotropic Gaussian linear regression ($\Sigma = I_d$) with balanced concepts and known concept assignments (oracle labels). The whitening reduction requires known Σ; extending to unknown, heterogeneous covariances is non-trivial. Our CIFAR-100 bridge is qualitative and tests only the high-SNR regime; a comprehensive real-data validation spanning both sides of the crossover remains future work. Extending the crossover characterization to nonlinear models, unbalanced concepts, and noisy concept inference are important open directions.

References

Hong-You Chen et al. Addressing concept shift in online federated learning. In *NeurIPS*, 2024.

Yuyang Deng, Mohammad Mahdi Kamani, and Mehrdad Mahdavi. Adaptive personalized federated learning. *arXiv preprint arXiv:2003.13461*, 2020.

Bradley Efron and Carl Morris. Stein's estimation rule and its competitors—an empirical Bayes approach. *Journal of the American Statistical Association*, 68(341):117–130, 1973.

Avishek Ghosh, Jichan Chung, Dong Yin, and Kannan Ramchandran. An efficient framework for clustered federated learning. In *NeurIPS*, 2020.

William James and Charles Stein. Estimation with quadratic loss. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, 1:361–379, 1961.

Ellango Jothimurugesan, Kevin Hsieh, Jianyu Wang, Gauri Joshi, and Phillip B Gibbons. Federated learning under distributed concept drift. In *AISTATS*, 2023.

Yoojin Kim et al. Communication-efficient personalized federated learning for speech-to-intent classification. In *ICML*, 2024.

Tian Li, Shengyuan Hu, Ahmad Beirami, and Virginia Smith. Ditto: Fair and robust federated learning through personalization. In *ICML*, 2021.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *AISTATS*, 2017.

Kunjal Panchal, Sunav Choudhary, Subrata Mitra, Koyel Mukherjee, Somdeb Sarkhel, Saayan Mitra, and Hui Guan. Flash: Concept drift adaptation in federated learning. In *ICML*, 2023.

Felix Sattler, Klaus-Robert Müller, and Wojciech Samek. Clustered federated learning: Model-agnostic distributed multitask optimization under privacy constraints. *IEEE TNNLS*, 2021.

Canh T Dinh, Nguyen Tran, and Josh Nguyen. Personalized federated learning with Moreau envelopes. In *NeurIPS*, 2020.

Alexandre B Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.

Rui Xu et al. Personalized federated learning with inferred collaboration graphs. In *ICML*, 2024.

A Proof of Proposition 1

Global estimator bias. Under balanced concepts with $\Sigma = I_d$, pooling all Kn samples mixes data from C regression problems $y = \langle w_j^*, x\rangle + \epsilon$. The population OLS solution on this mixture minimizes $\mathbb{E}[(y - \langle w, x\rangle)^2]$ over the mixture distribution $\frac{1}{C}\sum_j P_j$. Since $\mathbb{E}[xy] = \frac{1}{C}\sum_j w_j^* = \bar{w}^*$ and $\mathbb{E}[xx^\top] = I_d$, the population minimizer is $\bar{w}^*$, not w_j^*.

Global estimator excess risk. The excess risk for concept j decomposes as: $\mathbb{E}[\|\hat{w}_{\text{glob}} - w_j^*\|^2] = \|\bar{w}^* - w_j^*\|^2 + \mathbb{E}[\|\hat{w}_{\text{glob}} - \bar{w}^*\|^2] = B_j^2 + \mathbb{E}[\|\hat{w}_{\text{glob}} - \bar{w}^*\|^2]$. The cross-term $2(\bar{w}^* - w_j^*)^\top \mathbb{E}[\hat{w}_{\text{glob}} - \bar{w}^*]$ vanishes since $\mathbb{E}[\hat{w}_{\text{glob}}] = \bar{w}^*$. The variance term is $\mathbb{E}[\|\hat{w}_{\text{glob}} - \bar{w}^*\|^2] = \sigma^2 \text{tr}((X^\top X)^{-1})$. By random matrix theory, $\frac{1}{Kn}X^\top X \to I_d$ as $n/d \to \infty$, so $\text{tr}((X^\top X)^{-1}) = d/(Kn)(1 + O(d/(Kn)))$.

Concept-level estimator. Using only the $(K/C)n$ samples from $\mathcal{C}_j$, the OLS estimator $\hat{w}_j$ is unbiased for w_j^*. By the same argument, $\mathbb{E}[\|\hat{w}_j - w_j^*\|^2] = \sigma^2 d/((K/C)n)(1 + O(d/((K/C)n)))$.

B Proof of Theorem 3

Part 1 (Global bottleneck): For any single estimator $\hat{w}$ (possibly random), the average excess risk satisfies:

$$\begin{aligned}\frac{1}{C}\sum_{j=1}^{C}\|\hat{w} - w_j^*\|^2 &= \frac{1}{C}\sum_j \|(\hat{w} - \bar{w}^*) + (\bar{w}^* - w_j^*)\|^2 \\ &= \|\hat{w} - \bar{w}^*\|^2 + \frac{2}{C}\sum_j (\hat{w} - \bar{w}^*)^\top(\bar{w}^* - w_j^*) + \frac{1}{C}\sum_j \|w_j^* - \bar{w}^*\|^2.\end{aligned}$$

The cross-term vanishes because $\sum_j(\bar{w}^* - w_j^*) = 0$ by the definition $\bar{w}^* = \frac{1}{C}\sum_j w_j^*$. Hence $\frac{1}{C}\sum_j \|\hat{w} - w_j^*\|^2 = \|\hat{w} - \bar{w}^*\|^2 + \frac{1}{C}\sum_j \|w_j^* - \bar{w}^*\|^2 \geq B^2$. This is a deterministic inequality holding for *any* $\hat{w}$—even the infeasible choice $\hat{w} = \bar{w}^*$—and establishes that no single estimator can avoid the average bias B^2.

Part 2 (Concept-level bottleneck): The concept-level estimator $\hat{w}_j$ uses $(K/C)n$ i.i.d. Gaussian samples with $\Sigma = I_d$. By the standard minimax lower bound for d-dimensional linear regression [Tsybakov, 2009, Theorem 2.5], $\inf_{\hat{w}_j}\sup_{w_j^*}\mathbb{E}[\|\hat{w}_j - w_j^*\|^2] \geq \sigma^2 d/(2(K/C)n)$.

C Extended Experimental Results

Table 3 provides results for selected configurations from the full 108-setting sweep.

Table 3: Extended results (selected configurations). All MSE values are seed-averaged.

K	C	δ	τ	SNR	FedAvg	Oracle	Shrinkage	IFCA	CFL	APFL
10	2	0.3	∞	1.1	1.067	1.091	1.067	1.091	1.143	**1.065**
10	4	1.0	15	12.5	1.197	1.159	**1.155**	1.159	1.218	1.173
10	4	3.0	∞	113	3.657	**1.159**	1.159	1.159	1.225	1.286
20	8	1.0	15	12.5	1.235	1.231	**1.158**	1.231	1.303	1.224
40	4	3.0	5	450	3.694	1.081	**1.081**	1.081	1.114	1.159
40	8	8.0	15	1600	25.74	**1.066**	1.066	1.066	1.072	1.116

D Baseline Implementation Details

IFCA. Each client evaluates the MSE of all C cluster heads on its local data, selects the best, trains locally, and uploads. The server averages uploads per cluster. This is a linear-Gaussian analogue of Ghosh et al. [2020]; the original IFCA uses gradient descent, which we replace with closed-form OLS.

CFL. After one round of global OLS, we compute pairwise cosine similarity of client gradient updates $\nabla_k = \hat{w}_k^{\text{local}} - \hat{w}^{\text{global}}$. Clients are clustered hierarchically with C clusters. Each client is served by its cluster's averaged model. This is a linear-Gaussian analogue of Sattler et al. [2021].

APFL. For each $\alpha \in \{0.2, 0.5, 0.8\}$, we compute $\hat{w}_k^{\text{apfl}} = \alpha \hat{w}_k^{\text{local}} + (1-\alpha)\hat{w}^{\text{global}}$ and evaluate MSE. We report the best α per configuration. This is a tuned linear-Gaussian analogue of Deng et al. [2020].